

Date	03 July 2026
Team ID	SWTID-2026-4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
Team Member 2	Akili Venkata Siva Sai Srikar
Project Name	Machine Learning–Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Brainstorming & Idea Prioritization

1. Brainstorming

The project team conducted a structured brainstorming session to identify real-world problems that can be effectively solved using Artificial Intelligence and Machine Learning. Each proposed idea was evaluated based on its practical applicability, technical feasibility, data availability, business impact, implementation complexity, innovation potential, and long-term usefulness. After detailed discussion and comparison, the team selected the most suitable project for development.

2. Brainstormed Ideas

S.No	Project Idea	Problem Solved	Technical Feasibility	Business Impact	Priority
1	Machine Learning–Based Credit Card Approval Prediction System	Predicts whether a credit card application is likely to be approved or rejected using applicant information and machine learning models.	High	High	1
2	Student Performance Prediction System	Predicts academic performance using attendance, internal marks, and learning behaviour.	High	Medium	2
3	Employee Attrition Prediction System	Predicts whether employees are likely to leave an organization.	Medium	High	3
4	Loan Default Prediction System	Predicts the likelihood of loan repayment default.	High	High	4
5	Medical Disease Prediction System	Predicts possible diseases based on patient symptoms and clinical information.	Medium	High	5

3. Idea Evaluation Criteria

Evaluation Criterion	Importance	Credit Card Approval Prediction System
Real-world Problem	Very High	Excellent
Data Availability	High	Excellent
Machine Learning Suitability	Very High	Excellent
Implementation Complexity	Medium	Moderate
Business Value	Very High	Excellent
Innovation	High	Very Good
Scalability	High	Excellent
Future Enhancement	Very High	Excellent

4. Idea Prioritization

After evaluating all shortlisted ideas against the predefined selection criteria, the Machine Learning–Based Credit Card Approval Prediction System was selected as the final project. The project addresses a significant real-world problem faced by financial institutions by automating the preliminary credit card approval process. It offers high business value, strong applicability of Machine Learning algorithms, availability of suitable datasets, excellent scalability, and opportunities for future enhancement.

5. Reasons for Selecting the Project

- Solves a real-world banking problem.
- Reduces manual verification effort.
- Provides faster credit approval prediction.
- Improves consistency in decision making.
- Uses Artificial Intelligence and Machine Learning effectively.
- Has good dataset availability for model training.
- Supports scalable web-based implementation.
- Can be enhanced with explainable AI and advanced prediction models.

Conclusion
The brainstorming and idea prioritization process ensured that the selected project aligns with academic objectives, technical feasibility, industrial relevance, and business value. Based on a systematic evaluation, the Machine Learning–Based Credit Card Approval Prediction System was identified as the most suitable project for development due to its practical applicability, intelligent automation capabilities, and potential to improve credit card approval decision-making.